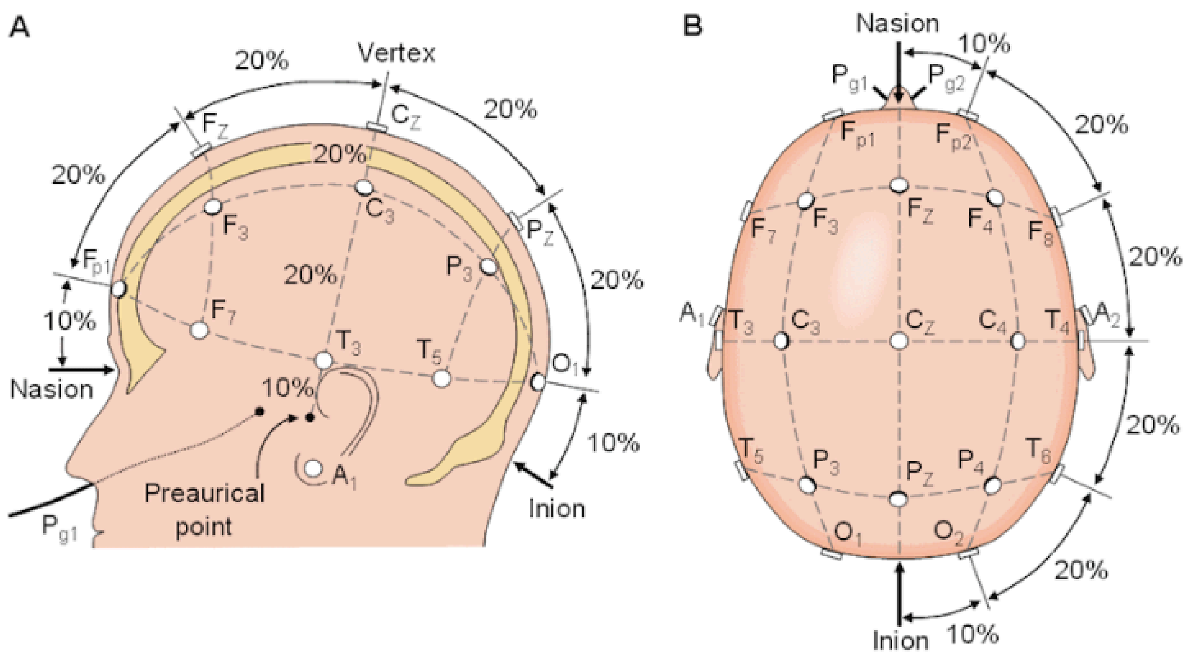
Meet Shah - 24116051 (EC2)

Electroencephalography (EEG)

EEG is a non-invasive method used to record the electrical activity of the brain. Electrodes are placed on the scalp to detect the synchronous firing of neurons, providing high temporal resolution that allows researchers to see brain changes in milliseconds.



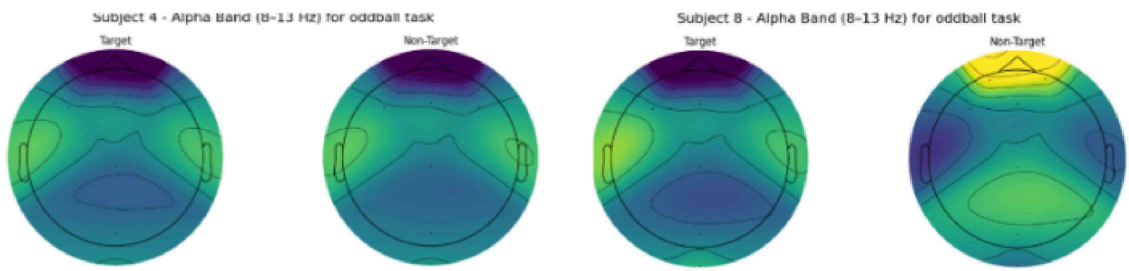
Brain Signal Bands and Their Significance

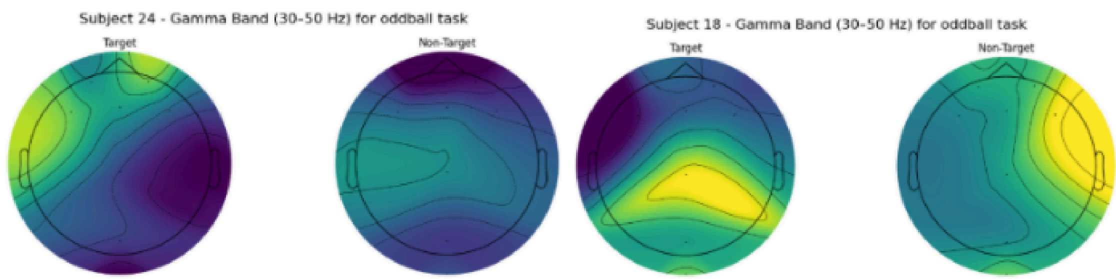
EEG data is typically categorized into different frequency bands, each associated with specific cognitive or physiological states:

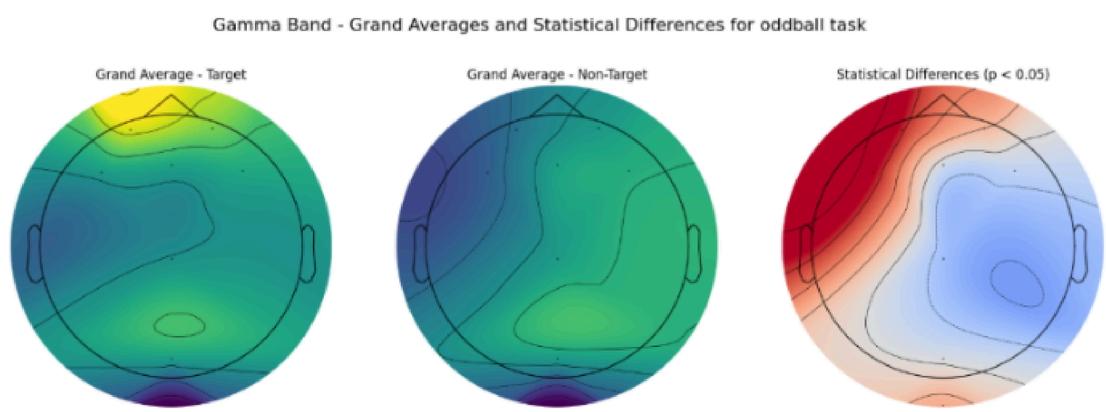
- **Delta (< 4 Hz)**: Dominant during deep, dreamless sleep.
- **Theta (4–8 Hz)**: Associated with drowsiness, light sleep, and memory processing.
- **Alpha (8–13 Hz)**: Occurs during relaxed wakefulness with eyes closed; it is often used to study attention and inhibition.
- **Beta (13–30 Hz)**: Linked to active thinking, focus, and motor behavior.
- **Gamma (> 30 Hz)**: Associated with high-level information processing and cognitive integration.

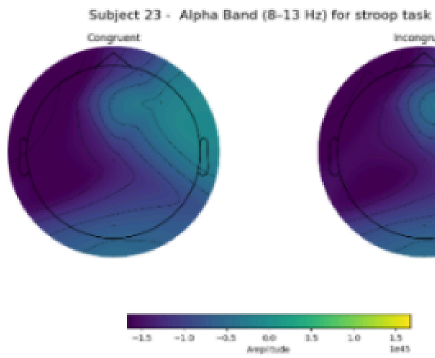
Cognitive Paradigms in EEG Research

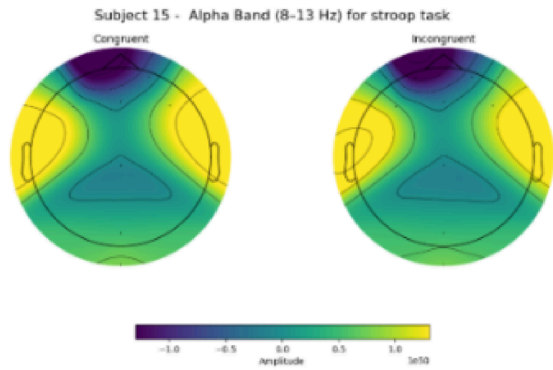
- **Oddball Paradigm**: A task where the subject is presented with a frequent standard stimulus and an infrequent target stimulus. It is primarily used to measure attention and the brain's ability to detect novelty (often associated with the P300 component).
- **Stroop Task**: Subjects name the color of a word (e.g., the word Blue printed in red ink). This measures cognitive interference and executive control.
- **Task-Switching Paradigm**: Requires subjects to switch between different rules or tasks rapidly, measuring cognitive flexibility and switch costs.
- **Dual-Task Paradigm**: Involves performing two tasks simultaneously to evaluate attentional resources and the brain's processing capacity limits.

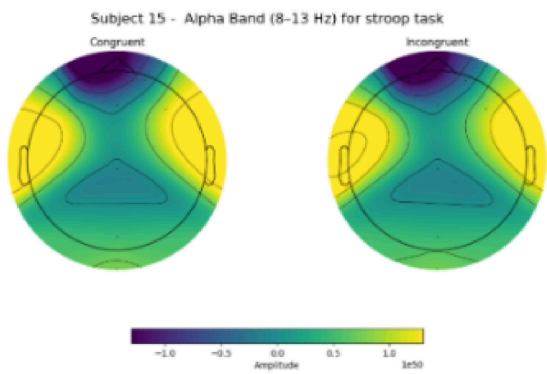


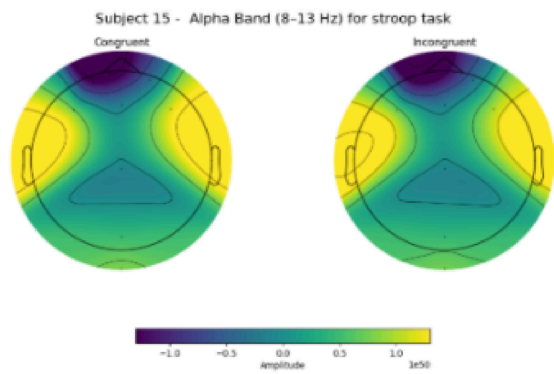


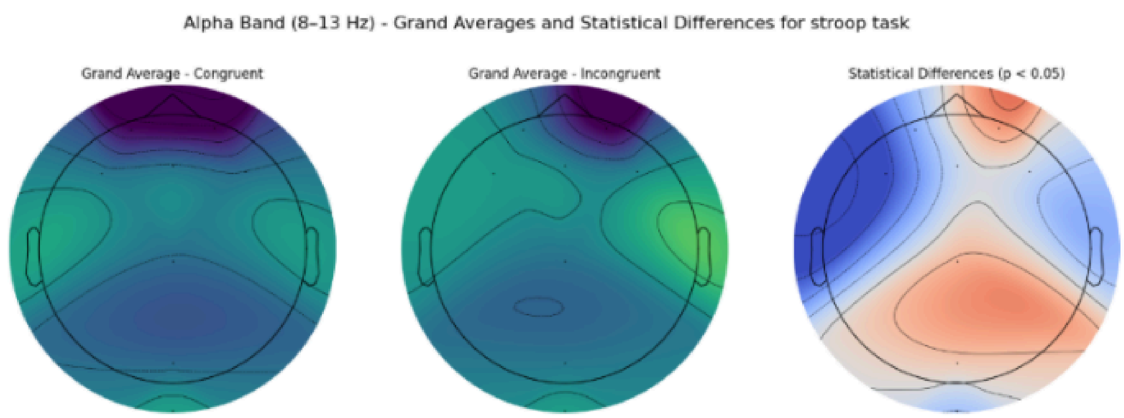












Analysis of Class Code

Our class code shows a complete pipeline for processing EEG data from multiple subjects (1 through 24) using the Python MNE library.

1. Data Preprocessing and Cleaning

The code implements a broad bandpass filter from 0.02 Hz to 35 Hz . This is a crucial step to remove low-frequency drifts (like sweating or movement) and high-frequency noise (like muscle activity or electrical interference).

2. Feature Isolation: The Alpha Band

A core component of this analysis is the specific isolation of the Alpha band (8–13 Hz). By applying a second filter restricted to this range, the analysis focuses on how relaxed or inhibited the brain state is during the task.

3. Epoching and Event Segmentation

The data is segmented into epoch, small windows of time relative to the stimuli. Our class code defines these from -0.2s to 0.8s around the stimulus onset . By averaging these epochs separately for Target and Non-Target stimuli, the code isolates the brain's specific response to the rare oddball stimulus .

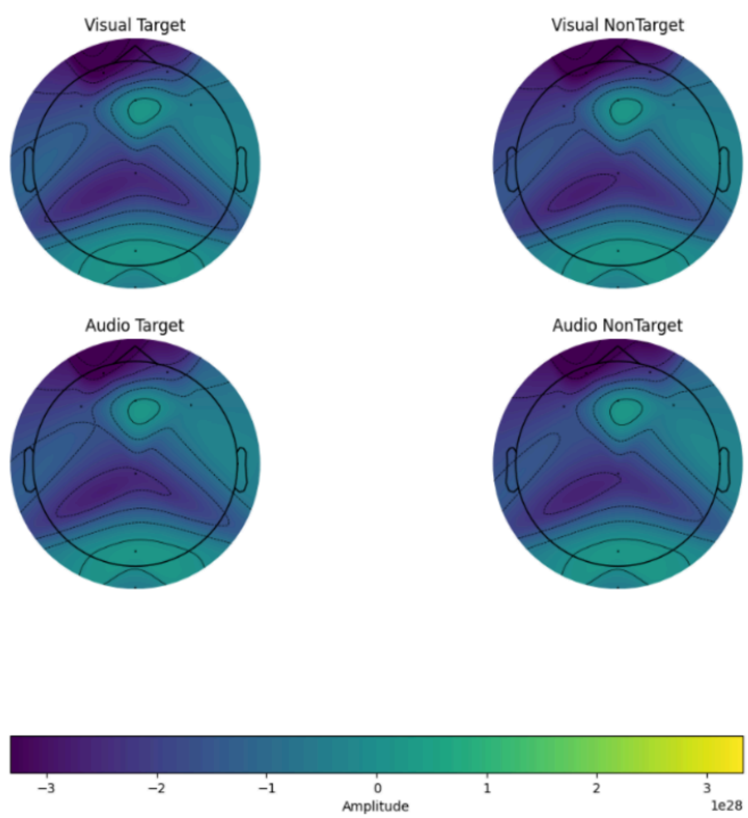
4. Statistical Validation

To ensure the findings are robust across the population, our class code a Grand Average analysis. This determines if the differences in alpha activity between the two conditions are statistically significant across all 24 subjects.

Interpretation of Outputs and Visualizations

Subject-Wise Topomaps

The topographical maps (topoplots) for individual subjects (e.g., Subject 1, 2, and 3) visualize the spatial distribution of alpha-band amplitude across the scalp.



- **Target vs. Non-Target**: In several subjects, the Target condition shows distinct changes in amplitude distribution compared to the Non-Target condition.
- **Alpha Suppression**: Often, the Target stimulus causes a localized decrease in alpha power (desynchronization) in parietal or frontal regions, indicating that the brain has moved from a relaxed state to an attentive state to process the rare stimulus.

Grand Average and Statistical Differences

The most critical output is the Grand Average Statistical Differences map.

- **Significance Mask**: Our class code uses a mask to highlight regions where the p-value is less than 0.05.
- **Spatial Significance**: The resulting graph shows specific clusters (often in the central or parietal sensors) where the brain significantly differentiates between the rare target and the frequent background.

Conclusions on Signal Band Effects

Based on our class code's outputs, we can conclude that the Alpha band is highly sensitive to the Oddball Paradigm. While standard EEG analysis often looks at raw voltage (ERPs), isolating the alpha band reveals that the brain's resting rhythm is modulated by cognitive demand.

The significant differences observed in the grand average plots confirm that the detection of a target stimulus disrupts the alpha rhythm in specific cortical regions. This demonstrates that the alpha band is not just a background signal but a dynamic component of the brain's attentional network.